WORKED SOLUTION

STEP I Specimen paper, Question 1

Pure • Discriminants and Partial Fractions

STEP I Specimen paper, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q1

The key idea

In part (i) the equation is quadratic in y for each fixed x, and a real y exists only when the discriminant is non-negative; that condition is exactly a restriction on x. Swapping the roles, the equation is also quadratic in x, which gives the restriction on y. Part (ii) is a standard partial-fractions question: split the fraction into pieces whose binomial expansions we know, then add the coefficients of x^n.

Solution

Part (i)

Regard the equation as a quadratic in y with coefficients depending on x:

$$y^2 + 16x\,y + (4x^2 + 24x) = 0.$$

For a given real x, there is a real y satisfying this if and only if the discriminant is non-negative:

$$\begin{aligned}(16x)^2 - 4(4x^2 + 24x) &= 256x^2 - 16x^2 - 96x \\ &= 240x^2 - 96x = 48x(5x - 2) \geq 0.\end{aligned}$$

The product $x(5x-2)$ is non-negative exactly when both factors have the same sign (or one is zero), that is, when $x \leq 0$ or $x \geq \frac{2}{5}$. So any real solution has

$$x \leq 0 \quad \text{or} \quad x \geq \tfrac{2}{5},$$

as required.

Now treat the equation as a quadratic in x:

$$4x^2 + (16y + 24)x + y^2 = 0.$$

A real x exists if and only if

$$\begin{aligned}(16y+24)^2 - 16y^2 &= 256y^2 + 768y + 576 - 16y^2 \\ &= 240y^2 + 768y + 576 \\ &= 48(5y^2 + 16y + 12) = 48(5y+6)(y+2) \geq 0.\end{aligned}$$

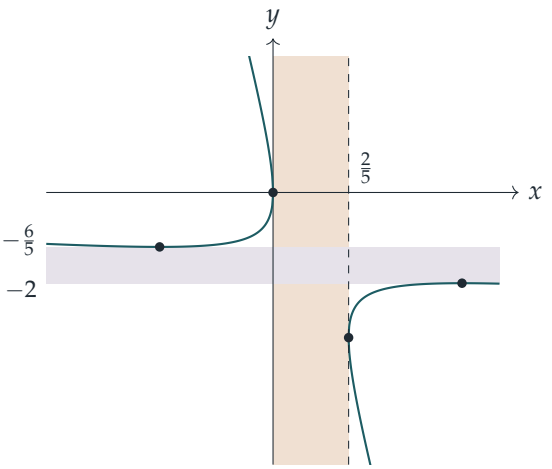
The roots of $(5y+6)(y+2)$ are $y = -2$ and $y = -\frac{6}{5}$, and the quadratic is negative only between them. Hence

$$y \leq -2 \quad \text{or} \quad y \geq -\tfrac{6}{5}.$$

Ask yourself Are these restrictions the best possible, or could the true ranges be smaller?

They are exact. The discriminant argument is an "if and only if": every x with $x \leq 0$ or $x \geq \frac{2}{5}$ gives at least one real y, and likewise for y. The boundary values are attained too, by the double roots: $(x, y) = (0, 0)$ and $(\frac{2}{5}, -\frac{16}{5})$ for the x-bounds, and $(1, -2)$ and $(-\frac{3}{5}, -\frac{6}{5})$ for the y-bounds. You can check each satisfies the original equation.

Geometrically the curve is a hyperbola (since $16^2 > 4 \cdot 4 \cdot 1$), and the two excluded strips are the gaps between its branches.



Part (ii)

Write

$$\frac{9}{(2-x)^2(1+x)} = \frac{A}{1+x} + \frac{B}{2-x} + \frac{C}{(2-x)^2},$$

so that $9 = A(2-x)^2 + B(2-x)(1+x) + C(1+x)$. Putting $x = -1$ gives $9 = 9A$, so $A = 1$; putting $x = 2$ gives $9 = 3C$, so $C = 3$; putting $x = 0$ gives $9 = 4A + 2B + C = 7 + 2B$, so $B = 1$. Thus

$$\frac{9}{(2-x)^2(1+x)} = \frac{1}{1+x} + \frac{1}{2-x} + \frac{3}{(2-x)^2}.$$

Now expand each piece, taking out the factor 2 so that the binomial series is in powers of $\frac{x}{2}$:

$$\frac{1}{1+x} = \sum_{n\geq 0}(-1)^n x^n, \qquad |x|<1,$$

$$\frac{1}{2-x} = \frac{1}{2}\Big(1-\frac{x}{2}\Big)^{-1} = \sum_{n\geq 0}\frac{x^n}{2^{n+1}}, \qquad |x|<2,$$

$$\frac{3}{(2-x)^2} = \frac{3}{4}\Big(1-\frac{x}{2}\Big)^{-2} = \frac{3}{4}\sum_{n\geq 0}(n+1)\frac{x^n}{2^n} = \sum_{n\geq 0}\frac{3(n+1)}{2^{n+2}}x^n, \qquad |x|<2,$$

using $(1-u)^{-2} = \sum_{n\geq 0}(n+1)u^n$ for $|u|<1$. Adding, the coefficient of x^n is

$$(-1)^n + \frac{1}{2^{n+1}} + \frac{3(n+1)}{2^{n+2}} = (-1)^n + \frac{2+3n+3}{2^{n+2}} = (-1)^n + \frac{3n+5}{2^{n+2}}.$$

All three series converge together exactly when $|x|<1$ (the first needs $|x|<1$, the others only $|x|<2$). For $1\leq|x|<2$ the last two still converge but the first does not, so their sum cannot converge either. So the expansion is valid for

$$-1<x<1.$$

Checks

In part (i), the point $(1,-2)$ gives $4-32+4+24=0$, and $(-\frac{3}{5},-\frac{6}{5})$ gives $\frac{36}{25}+\frac{288}{25}+\frac{36}{25}-\frac{360}{25}=0$, confirming that both y-bounds are attained. In part (ii), $n=0$ gives $1+\frac{5}{4}=\frac{9}{4}$, which matches $\frac{9}{4\cdot 1}$ at $x=0$; and $n=1$ gives $-1+\frac{8}{8}=0$, which is right because the denominator $(2-x)^2(1+x)=4-3x^2+x^3$ has no x term. Numerically, summing the series at $x=0.9$ gives $3.914745\ldots$, agreeing with $9/(1.1^2\times 1.9)$.

Common slips

- Getting the inequality the wrong way round and answering $-2\leq y\leq -\frac{6}{5}$. The discriminant is negative *between* its roots, so that interval is the one excluded (and $(0,0)$ lies on the curve with $y=0$ outside it).
- Expanding $(2-x)^{-1}$ as if it were $(1-x)^{-1}$, forgetting to take out the factor $\frac{1}{2}$ (or $\frac{1}{4}$ for the squared term).
- Giving the range of validity as $|x|<2$: the expansion is only valid where every piece converges, so $|x|<1$.

Beyond STEP

The curve in part (i) is a conic, and whether it is an ellipse, parabola or hyperbola depends on the sign of b^2-4ac for its quadratic terms $ax^2+bxy+cy^2$. At university

this is seen through the symmetric matrix of the quadratic form and its eigenvalues. Part (ii) touches the idea of a radius of convergence: the expansion is valid out to the nearest point where the function blows up, here $x = -1$. Strang's *Introduction to Linear Algebra* covers quadratic forms, and Abbott's *Understanding Analysis* covers power series and their convergence.

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